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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: B.Tech Stream: CSBS | Year: II Semester: III

Subject: Object Oriented Programming | Time: 03 hrs (11:00 am to 2:00 pm)

Date: 14 / 06 / 23 | No. of Pages: 2

Marks: 100

Special Re- Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1 BL-2	a.	Explain the concept of structures with the help of an example	5M
CO- 1 BL-2	b.	Why do we need objects in object-oriented programming? Justify your answer with example	5 M
CO-2 BL-1	c.	List Which unary operators can be overloaded? Write a small code snippet for any one operator explaining the same.	5 M
CO-4 BL-2	d.	Explain the components of Use Case Diagram with the help of an example	5 M
Q2 CO-1 BL-6	a	Write a program to implement following: Create structure to manage the student data. • Name of the Student • Student SAP Number • CGPA of Student • Branch of Student Implement following functions to perform operations 1. Read the data of 2 Students 2. Display the data of Students 3. Search the specified Student by name, if it is available or not, if available then display student record. 4 Search the specified Student by SAP number, if it is available or not, if available then display employee record	[10]
CO-3 BL-2	b	Explain the need of functions in procedural oriented Programming. Explain default arguments in function with the help of an example	[10]

Q3 CO-2 BL-3	a	What is the need of Inheritance? Explain different types of inheritance	[10]
CO-3 BL-3	b	Explain the concept of Template class and template function with the help of an example.	[10]
Q4 CO-2 BL-2, 6	a	What is the need of Friend Function? Justify your answer with the help of an example	[10]
CO-4 BL-6	b	Design class diagram for Movie Ticket Booking system with detailed explanation of each component of class diagram	[10]
Q5 CO-2 BL-3	a	Write a Program demonstrating the concept of Call by Value and Call by Reference	[10]
CO-4 BL-6	b	Design Activity diagram from Bank Management System with swim lanes	[10]
Q6 CO-2 BL-6	a	Explain the concept of classes and object with the help of an example	[10]
CO-2 BL-1 2 6	b	Why are constructors called specific member functions? Explain the use of constructors in object-oriented programming with the help of an example	[10]
Q7 CO-2 BL-2	a	Explain Function Overriding with the help of an example	[5]
CO-2 BL-2	b	Define abstraction. Explains it with a real time example.	[5]
CO2, BL3.	c	Write a program depicting the concept of function overloading.	[5]
CO-2 BL-2	d	Explain the use of this keyword with the help of an example.	[5]